

DCN MODULE 01

Question Bank For Module 01 (By Me)

Q1 What is Data, Data communication?

Data refers to the raw facts that are collected while information refers to processed data that enables us to take decisions. Data communications refers to the transmission of this digital data between two or more computers and a computer network or data network. Data Communication depends on

- Jitter
- Accuracy
- Timeliness
- Delivery

Q2 what is Data Representation?

A network connects multiple devices, enabling communication and resource sharing, such as printers and scanners. Data transferred in a network exists in various formats:

Text – Represented as bit patterns and converted using character encoding (e.g., ASCII, Unicode). (File formats: .txt, .pdf, .docx, etc.)

Numbers – Stored as binary bit patterns for mathematical operations. (Formats: Integer, Boolean, Decimal, etc.)

Images – Made up of pixels, stored in binary form. (Formats: JPEG, PNG, GIF, etc.)

Audio – Digital representation of sound waves, continuous in nature. (Formats: MP3, WAV, AAC, etc.)

Video – Sequence of images in motion, used for recording and broadcasting. (Formats: MP4, AVI, MOV, etc.)

Q3 What Does Network Mean?

A network, in computing, is a group of two or more devices or nodes that can communicate. The devices or nodes can be connected by physical or wireless connections.

Networks may be used for:

- ✓ Communications such as email, instant messaging, chat rooms, etc.
- ✓ Shared hardware such as printers and input devices.
- ✓ Shared data and information through the use of shared storage devices.
- ✓ Shared software, which is achieved by running applications on remote computers.

Benefits of Data Communication

- Connectivity

We can think of it along three types of connections:

- Person to Person
- Person to Machine
- Machine to Machine

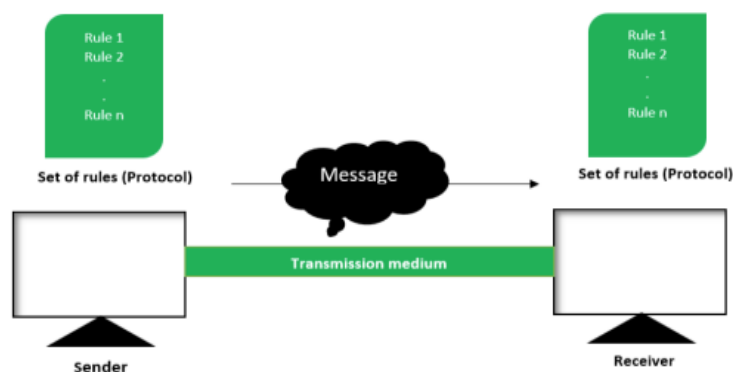
Q4 What are the components of the Data Communication System?

Data communication is the exchange of data between two or more devices through a transmission medium. It enables digital devices to share information efficiently.

Components of Data Communication System:

1. **Message** – It is the data or information to be transmitted, such as text, audio, video, or images. Communication is meaningless without a message.
2. **Sender** – The device that generates and transmits the message, such as a computer, mobile phone, or server. It acts as the source of data.
3. **Receiver** – The device that accepts and processes the transmitted message. It can be any digital device like a computer, phone, or printer.
4. **Transmission Medium** – The physical or wireless channel through which data travels from sender to receiver. Examples include fiber optic cables, radio waves, and microwaves.

5. **Protocol** – A set of rules that define how data is transmitted and interpreted. Without a protocol, communication between devices would be impossible.

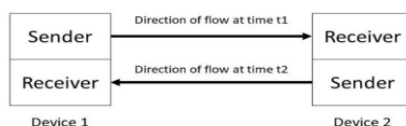
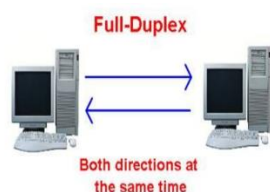


Q5 What is Data Flow? Simplex, Half-Duplex, Full-Duplex?

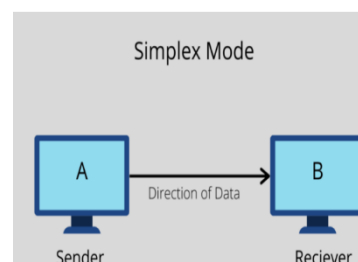
In networking, data flows between devices using different transmission modes. These modes define how information moves from one device to another. The three primary types of data communication are **Simplex**, **Half-Duplex**, and **Full-Duplex**.

1. Simplex Communication

- ◆ **Definition:** One-way (unidirectional) communication where one device sends data and the other only receives it. The receiver cannot send any response.
- ◆ **Key Feature:** The entire bandwidth is used for transmission in a single direction.
- ◆ **Examples:**
 - TV broadcast (TV receives signals from the station but cannot send back).
 - Keyboard input (Keyboard sends data to the computer; the computer doesn't send data back to the keyboard).
 - Radio stations (You can only listen, not talk back).



Half-Duplex Mode



2. Half-Duplex Communication

- ◆ **Definition:** Two-way (bidirectional) communication, but only one device can send data at a time while the other receives.
- ◆ **Key Feature:** Devices take turns to communicate, meaning there is a delay in response.
- ◆ **Examples:**
 - Walkie-talkies (Only one person can speak at a time, and the other listens).
 - Two-way radios used in security communication.

3. Full-Duplex Communication

- ◆ **Definition:** Two-way (bidirectional) communication where both devices can send and receive data simultaneously.
- ◆ **Key Feature:** No waiting time for response, making it the fastest and most efficient mode.
- ◆ **Examples:**
 - Mobile phones (You can talk and listen at the same time).
 - Landline telephones.
 - Video calls (Both users can see and hear each other at the same time).

Q6 what is networking devices? (2m)

Network Devices: Network devices, also known as networking hardware, are physical devices that allow hardware on a computer network to communicate and interact with one another. For example Repeater, Hub, Bridge, Switch, Routers, Gateway and NIC, etc.

Q7 Define a computer network. What are the four main types of computer networks?

A **computer network** is a system where multiple computers and devices are connected to share data, resources, and applications. Based on size and coverage, networks are categorized into four types: **LAN, PAN, MAN, and WAN**.

1. Local Area Network (LAN)

- Covers a small area like a home, office, or school.
- Uses inexpensive hardware like hubs, switches, and Ethernet cables.
- Provides high-speed data transfer.
- Ensures better security than larger networks.

2. Personal Area Network (PAN)

- Connects personal devices within a 10-meter range (e.g., Bluetooth, USB).
- First introduced by Thomas Zimmerman.
- Can be **wired (USB connections)** or **wireless (WiFi, Bluetooth)**.

3. Metropolitan Area Network (MAN)

- Covers a city or metropolitan area by interconnecting multiple LANs.
- Used by government agencies, banks, airlines, and colleges.
- Uses technologies like **ISDN, ADSL, and Optical Carriers (OC-3)**.

4. Wide Area Network (WAN)

- Spans large geographical areas, like countries or continents.
- The **Internet is the largest WAN**.
- Used in businesses, governments, and education.
- Connects locations using **fiber optics, satellite, and leased lines**.

5. Advantages of WAN

1. **Large Coverage:** Connects distant locations worldwide.
2. **Centralized Data:** No need for multiple backup servers.
3. **Fast Communication:** Supports web apps like WhatsApp & Skype.
4. **Resource Sharing:** Allows sharing of software and storage.
5. **High Bandwidth:** Leased lines improve data transfer speed.

6. Disadvantages of WAN

1. **Security Risks:** More vulnerable to hacking and cyber threats.
2. **Firewall & Antivirus Needed:** Protects against viruses and attacks.
3. **High Setup Cost:** Requires routers, switches, and leased lines.
4. **Troubleshooting Issues:** Harder to fix network failures over large areas.

Q8. What is Network Topology? – Physical Structures (10 Marks)

Definition

- Network topology refers to the physical or logical arrangement of devices in a network.
- Defines how computers and other devices are interconnected for communication.
- Physical topology deals with actual connections, while logical topology defines data flow.

Types of Network Topologies

1. Point-to-Point Topology

- Direct connection between two devices (sender and receiver).
- Provides high bandwidth and direct communication.
- Simple structure with minimal configuration.
- Used in leased lines, telecommunication links, and direct computer connections.

2. Bus Topology

- All devices share a single backbone cable.
- Data is transmitted along the backbone, and all nodes receive it.
- Uses **CSMA (Carrier Sense Multiple Access)** for transmission.
- Requires terminators at both ends to prevent signal reflection.
- Suitable for small networks but prone to failures if the backbone is damaged.
- Examples: Early Ethernet (10Base2, 10Base5).

3. Ring Topology

- Devices are connected in a circular fashion.
- Data flows in one direction (unidirectional) using **token passing**.
- No data collisions since only one device transmits at a time.
- Failure in one node disrupts the network unless a dual-ring system is used.
- Example: Token Ring networks, Fiber Distributed Data Interface (FDDI).

4. Star Topology

- All devices connect to a **central hub or switch**.
- The hub acts as a communication relay between devices.
- **Easier troubleshooting** as failure in one node does not affect others.
- Hub failure causes complete network failure.
- Widely used in Ethernet LANs.

5. Tree Topology

- Hierarchical structure combining **bus and star topologies**.
- Central root node connects to other nodes, forming a **parent-child hierarchy**.
- Segments can be managed separately, making troubleshooting easier.
- Failure of the **main bus cable** affects the entire network.
- Common in **corporate networks and WANs**.

6. Mesh Topology

- Every device has multiple connections to other devices.
- **Full Mesh**: Each node connects to all other nodes.
- **Partial Mesh**: Only selected nodes have multiple connections.
- Provides **high reliability and redundancy** but is costly and complex.
- Example: The Internet, wireless mesh networks.

7. Hybrid Topology

- Combination of multiple topologies to suit specific network needs.
- Provides **scalability, reliability, and flexibility**.
- Complex **design and expensive infrastructure**.
- Used in **large enterprises and data centers**.

Comparison Table for Exam Perspective

Topology	Structure	Advantages	Disadvantages	Example
Point-to-Point	Direct connection	High speed, simple setup	Limited to two nodes	Direct cable connection between two routers
Bus	Single backbone cable	Cheap, easy to expand	Cable failure affects all	Early Ethernet
Ring	Circular loop	No data collisions, efficient	Single point failure	Token Ring networks
Star	Central hub	Easy troubleshooting, scalable	Hub failure affects all	Ethernet LANs
Tree	Hierarchical	Scalable, easy to manage	Expensive, complex	WANs, large organizations
Mesh	Fully interconnected	Reliable, fast	High cost, difficult setup	Internet, wireless mesh networks
Hybrid	Combination of topologies	Flexible, scalable	Complex, expensive	Enterprise networks

Q9 What is Network Criteria?

A Network is a group of connected devices capable of communicating. The device can be a computer, printer, or scanner that either receives or transmits data. The devices connected in the Network are called "Nodes," and the means of connection can be wired or wireless. There are a lot of criteria that make a network better than others, but; there are three basic yet important criteria to be fulfilled for a network:

- **Transit Time** – Total time taken for a message to be transmitted from start to end.
- **Response Time** – Time taken for a system to process a request and respond.
- **Throughput** – Amount of data successfully transmitted per unit time.
- **Bandwidth** – Maximum data capacity of the network.
- **Latency (Delay)** – Time delay in data transmission due to network factors.
- **Reliability** – Measured by failure frequency, recovery time, and robustness in disasters.
- **Security** – Protection against unauthorized access, attacks, and data breaches.

Q10 What are Network Models?

Layered Architecture in Computer Networks

- A **divide-and-conquer** approach where network processes are split into smaller sub-tasks.
- Each **layer** handles a specific task independently.
- **Lower layers** provide services to **higher layers** above them.
- **Modifications in one layer don't affect others**, making the system modular and flexible.

Key Elements of Layered Architecture

1. **Service:** Actions provided by a layer to the layer above.
2. **Protocol:** Set of rules for communication between peer entities.
3. **Interface:** Defines how layers interact and transfer messages.

Why Layered Architecture?

- **Reduces complexity** through modularity.
- **Easy to modify & test**, as each layer operates independently.
- **Efficient communication** across different network systems.

Network Models using Layered Architecture

1. **OSI Model (7 layers)** – Standardized model for network communication.
2. **TCP/IP Model (4 layers)** – Practical model used in the internet.

Interconnection Networks

Used for communication between processors in a network.

- **Components:** Links & switches help transfer data between nodes.
- **Goals:** Low latency, high concurrency, and cost-effectiveness.

Types of Interconnection Networks

1. **Static Interconnection Networks (Direct Networks)**
 - Fixed point-to-point links between processors.
 - **Used in centralized & distributed systems** where communication is predictable.
 - **Examples:** Linear Array, Ring, Mesh, Tree, Hypercube, Butterfly.
2. **Dynamic Interconnection Networks (Indirect Networks)**
 - Uses **switches & dynamic links** for flexible communication.
 - Can handle multiple communication patterns based on program demands.
 - **Examples:** Bus, Crossbar Network, Multistage Network, Omega Network.

Q11 What is the Internet?

- The **Internet** is a vast interconnected network of computers and devices that enables global communication and information sharing.
- It allows access to **websites, emails, social media, and online services** by connecting millions of computers and servers.
- The Internet was first developed in the **1960s** with **ARPANET**, which introduced **packet switching** for communication.
- In **1969**, the first message was sent using ARPANET, marking the foundation of modern networking.
- Today, the Internet is the **fastest medium** for transmitting data, supporting everything from messaging to cloud computing.

A network protocol is an accepted set of rules that govern data communication between different devices in the network. It determines what is being communicated, how it is being communicated, and when it is being communicated. It permits connected devices to communicate with each other, irrespective of internal and structural differences

Q12 Key Element of Protocol?

- **Syntax** – Defines the structure and format of exchanged data, including sender/receiver addresses and message composition.
- **Semantics** – Establishes rules for interpreting data, ensuring correct message understanding and response.
- **Timing** – Synchronizes data transfer, preventing data loss due to speed mismatches between sender and receiver.
- **Sequence Control** – Ensures data packets are received in the correct order and retransmits lost packets.
- **Flow Control** – Regulates data flow to prevent congestion and ensures the receiver can handle incoming data.
- **Error Control** – Detects and corrects transmission errors using error detection codes and retransmission methods.
- **Security** – Protects data confidentiality, integrity, and authenticity using encryption, authentication, and access control.

Q13 What are Different types of protocols?

1. Network Communication Protocols

These define the rules for data transfer across networks.

- **HTTP (HyperText Transfer Protocol):** Enables communication between web browsers and servers.
- **TCP (Transmission Control Protocol):** Ensures reliable, sequential data transmission without duplication.
- **IP (Internet Protocol):** Handles packet routing and addressing to ensure data reaches the correct destination.
- **UDP (User Datagram Protocol):** A faster but connectionless protocol used for broadcasts and streaming.
- **FTP (File Transfer Protocol):** Enables file sharing between servers with secure control mechanisms.

2. Network Security Protocols

These protect data from unauthorized access using encryption.

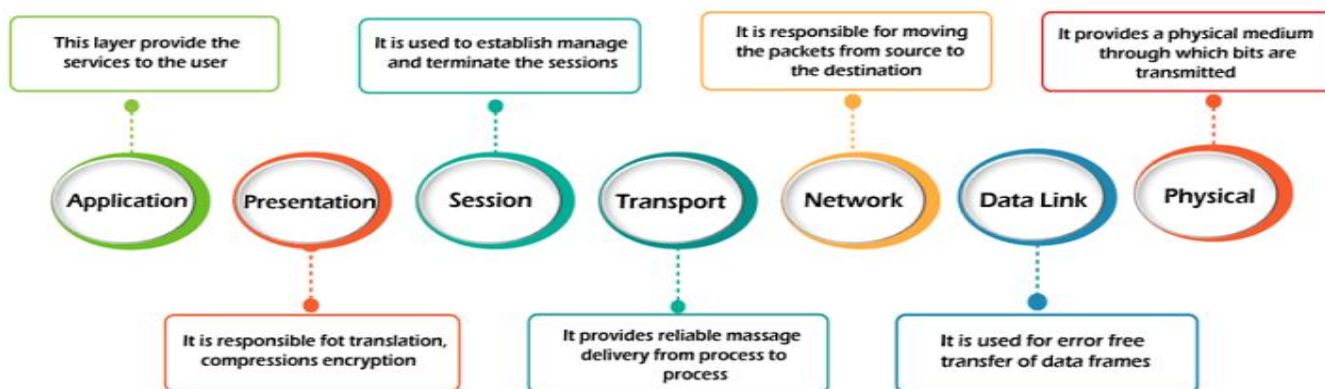
- **SFTP (Secure File Transfer Protocol):** Encrypts file transfers for security.
- **HTTPS (HyperText Transfer Protocol Secure):** A secure version of HTTP using encryption.
- **SSL (Secure Socket Layer):** Encrypts internet communications to protect sensitive data.

3. Network Management Protocols

These assist in monitoring and troubleshooting network performance.

- **SNMP (Simple Network Management Protocol):** Helps administrators manage and monitor network devices.
- **ICMP (Internet Control Message Protocol):** Diagnoses network connectivity issues, congestion, and timeouts.

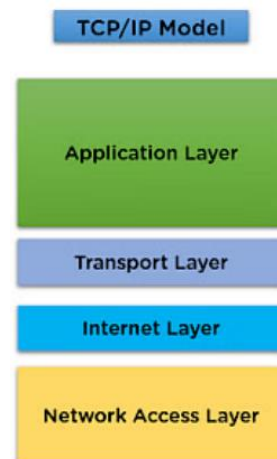
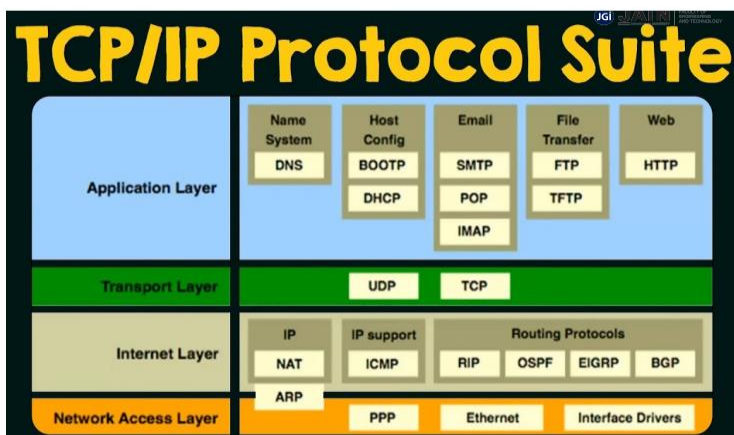
Q14 What is OSI Model? (10M Guaranteed Q)



Layer No	Layer Name	Responsibility	Information Form(Data Unit)	Device or Protocol
7	Application Layer	Helps in identifying the client and synchronizing communication.	Message	SMTP
6	Presentation Layer	Data from the application layer is extracted and manipulated in the required format for transmission.	Message	JPEG, MPEG, GIF
5	Session Layer	Establishes Connection, Maintenance, Ensures Authentication, and Ensures security.	Message	Gateway
4	Transport Layer	Take Service from Network Layer and provide it to the Application Layer.	Segment	Firewall
3	Network Layer	Transmission of data from one host to another, located in different networks.	Packet	Router
2	Data Link Layer	Node to Node Delivery of Message.	Frame	Switch, Bridge
1	Physical Layer	Establishing Physical Connections between Devices.	Bits	Hub, Repeater, Modem, Cables

- **Overall Concept:**
OSI stands for Open Systems Interconnection—a 7-layer reference model developed by ISO in 1984 to standardize network communication.
- **Layered Structure:**
The model breaks the complex task of networking into 7 distinct, manageable layers, each with its own responsibilities.
- **Physical Layer:**
The lowest layer, it handles raw bit transmission over physical media (cables, hubs, etc.), including bit synchronization, bit rate control, and specifying transmission modes (simplex, half-duplex, full-duplex).
- **Data Link Layer:**
This layer creates error-free communication between adjacent nodes by framing data, adding MAC addresses, and controlling access to the physical medium. It has two sublayers: the Logical Link Control (LLC) and Media Access Control (MAC).
- **Network Layer:**
Responsible for logical addressing and routing, it directs packets across networks using protocols like IP, ensuring data finds the most efficient path to its destination.
- **Transport Layer:**
Acts as the "heart" of data delivery by segmenting and reassembling messages, providing end-to-end error control, flow control, and supporting both connection-oriented (TCP) and connectionless (UDP) communications.
- **Session Layer:**
This layer establishes, manages, and terminates sessions between applications. It synchronizes communication (using checkpoints) and controls dialogues to keep data streams orderly.
- **Presentation Layer:**
Often called the translation layer, it formats, encrypts/decrypts, and compresses data, ensuring that information is in a usable form for the application layer.
- **Application Layer:**
The topmost layer where user-end applications reside. It provides network services directly to end-user applications like browsers, email clients, and file transfer programs.
- **Modular Benefits:**
The OSI model's layered design allows each layer to operate independently. This modularity simplifies troubleshooting, enhances flexibility, and permits upgrades without overhauling the entire system.

Q15 TCP/IP Model (Important)



- **Physical Layer :** It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver.
- **Data Link Layer :** TCP/IP, is identified by the data-link layer. Error prevention and “framing” are also provided by the data-link layer. Point-to-Point Protocol (PPP) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

- **Internet Layer :** The Internet Layer is responsible for routing packets of data from one device to another across a network. It does this by assigning each device a unique IP address, which is used to identify the device and determine the route that packets should take to reach it. It defines the protocols which are responsible for the logical transmission of data over the entire network.
 1. **IP (Internet Protocol):** Routes data packets using IP addresses.
 2. **ICMP (Internet Control Message Protocol):** Sends error and status messages to help diagnose network issues.
 3. **ARP (Address Resolution Protocol):** Translates IP addresses into physical (MAC) addresses.
 4. **TCP (Transmission Control Protocol):** Provides reliable, ordered data transfer, like a virtual circuit.
 5. **UDP (User Datagram Protocol):** Sends data quickly without ensuring delivery, ideal for real-time apps.
- **Application Layer :** The Application Layer provides end-to-end communication, hiding network complexity from user applications. It ensures data is delivered error-free and efficiently. Key protocols include:
 - 1) **HTTP/HTTPS:** HTTP manages web browsing, while HTTPS adds encryption for secure transactions.
 - 2) **SSH:** Enables secure, encrypted remote terminal access.
 - 3) **NTP:** Synchronizes clocks across systems to keep time-sensitive transactions accurate.
 - 4) Additionally, protocols like FTP handle file transfers and SMTP manages email communications.

Q16. What are Computer addressing?

Addressing Network Addressing is one of the major responsibilities of the network layer. Network addresses are always logical, i.e., software-based addresses. A host is also known as end system that has one link to the network. The boundary between the host and link is known as an interface

1. **Hardware Addressing (MAC Addresses):**
 - **Layer:** Performed at the Data Link layer (Layer 2) of the OSI model.
 - **Function:** Uniquely identifies hosts on a local network.
 - **Details:** Each Network Interface Card (NIC) has a unique, hard-coded MAC address.
 - **Size & Format:** 48 bits long, represented in hexadecimal (e.g., 06:5f:39:ac:dd:2c).
2. **Logical Addressing (IP Addresses):**
 - **Layer:** Managed at the Network layer of the OSI model.
 - **Function:** Uniquely identifies hosts across entire networks and separates different networks.
 - **Details:** IP addresses are assigned to interfaces, allowing data routing between networks.
 - **Size & Format:** Typically 32 bits long (IPv4) and represented in dot-decimal notation (e.g., 193.32.216.9). An IP address consists of a network ID (identifies the host's network) and a host ID (uniquely identifies the host within that network).
 - **Additional Info:** Logical addresses can be dynamic, allowing for reconfiguration and scalability. The Internet Protocol (IP) handles both logical addressing and routing.

Together, hardware addressing ensures local communication, while logical addressing enables global connectivity and efficient packet routing.

Q17 IEEE 802.11 Architecture? (Very Imp Q)

The **IEEE 802.11** architecture defines the standard for **Wireless Local Area Networks (WLANs)**, commonly known as Wi-Fi. It consists of various components that enable wireless communication.

1. Stations (STA)

- Devices connected to the wireless LAN, like laptops, smartphones, and printers.
- Each station has a **wireless network interface controller (NIC)**.

2. Types of Stations

- **Wireless Access Points (WAP):** Act as base stations for the network.
- **Clients:** Devices like laptops and mobile phones that connect to the network.

3. Basic Service Set (BSS)

- A **group of stations** communicating at the physical layer.
- **Infrastructure BSS:** Devices communicate via an **access point (AP)**.
- **Independent BSS:** Devices communicate **directly (peer-to-peer/ad hoc mode)**.

4. Extended Service Set (ESS)

- A set of multiple connected BSS forming a larger wireless network.

5. Applications of IEEE 802.11

- **Home Networking:** Connects laptops, smart TVs, and gaming consoles.
- **Wi-Fi Hotspots:** Provides free or paid internet in public places like malls and airports.
- **Campus Connectivity:** Offers network access in schools, universities, and offices.

6. Advantages of WLANs

1. **Wireless & clutter-free** – No need for cables.
2. **Scalable** – Devices can be added or removed easily.
3. **Portable** – Users can move freely within network range.
4. **Easy installation** – Simpler setup than wired networks.
5. **Cost-effective** – Reduces hardware and installation costs.

7. Disadvantages of WLANs

1. **Signal Interference** – More noise and interference from other devices.
2. **Security Risks** – Requires encryption to prevent hacking.
3. **More Errors** – Prone to data loss and requires higher bandwidth.
4. **Slower than Wired LANs** – Wireless connections have lower speeds.

Q18 Wireless Links and Network Characteristics?

- ✓ **Attenuation:** The reduction in signal strength as it travels through a medium, leading to weaker signals at the receiver.
- ✓ **Path Loss:** A specific form of attenuation where signal strength decreases with distance as electromagnetic waves disperse in free space or through obstacles.
- ✓ **Multipath Propagation:** Occurs when transmitted waves reflect off surfaces and arrive at the receiver via different paths, potentially causing interference and time delays.
- ✓ **CRC Error Detection:** A method using cyclic redundancy check codes to detect errors in transmitted data blocks, ensuring data integrity.
- ✓ **Signal-to-Noise Ratio (SNR):** A measure of signal quality expressed in decibels (dB) that compares the strength of the desired signal to the background noise.
- ✓ **Bit Error Rate (BER):** The probability that a transmitted bit is incorrectly received, indicating the overall quality of the transmission.
- ✓ **Modulation:** The process of converting digital data into electrical signals that are optimized for transmission over a specific medium.
- ✓ **Hidden Terminal Problem:** Occurs when physical obstructions prevent devices from detecting each other's transmissions, leading to potential data collisions at a common receiver.
- ✓ **Fading:** The variation or decrease in signal strength over time or distance as the signal propagates through the wireless medium.